

Naram Mhaisen

Curriculum Vitae

Delft, The Netherlands

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Date of birth: 22 May 1994

Residence status: EU long-term resident, Netherlands.

Research interests: Online learning, regret minimization, optimistic learning, and network optimization.

🔗 Google Scholar | 📄 ResearchGate | 🐙 GitHub | 🔗 LinkedIn

Education

04/2021– **Ph.D. in Computer Science**, *Delft University of Technology*, Netherlands.

- 04/2025 ○ Thesis: “Optimistic Learning with Applications to Caching Networks” 🔗
- Supervisors: George Iosifidis, Koen Langendoen.

09/2018– **Master of Science in Computing**, *Qatar University*, Qatar, *GPA: 3.88/4.00*.

- 06/2020 ○ Thesis: “Novel Techniques for Blockchain-enabled IoT leveraging Reinforcement Learning” 🔗
- Supervisors: Mohsen Guizani.
- “Distinguished Research” Award for obtaining >3.8 GPA & publishing the findings.

09/2013– **Bachelor of Science in Computer Engineering**, *Qatar University*, Qatar, *GPA: 4.00/4.00*.

- 08/2017 ○ “Order of Excellence” Award for obtaining >3.8 GPA.
- “Honors” Award for completing Qatar University Honors Program.

Experience

04/2025– **Postdoctoral Researcher**, *Delft University of Technology*, Netherlands

- Present ○ Research and Development (R&D) in a Horizon Europe project “ORIGAMI” for using online learning methods in resource allocations across networks.

07/2020– **Research Assistant**, *Qatar University*, Qatar

- 04/2021 ○ R&D in a “High Impact” project titled: “Edge computing for scalable and secure medical data exchange using Blockchain”.

10/2018– **Graduate Research/Teaching Assistant**, *Qatar University*, Qatar

- 07/2020 ○ R&D in applying reinforcement learning for the optimization of IoT systems.
- A member of the Cyber Range challenge team in cooperation with Thales.

09/2017– **Research Assistant**, *Qatar University*, Qatar

- 08/2018 ○ R&D at “National Priority Project”: “Efficiently reliable and privacy-preserving cloud storage”.

07/2017– **Software Engineer**, *Qatar Mobility Innovation Center (QMIC)*, Qatar

- 09/2017 ○ Hardware/Software design and implementation of a prepaid energy embedded system based on Ethereum Blockchain and IoT technologies.

Awards

2022 “Research Fellowship” Award, NEC Laboratories-Europe 🔗

2020 “Best Demo Award” IEEE Inter. Conference for Informatics, IoT, and Enabling Tech

2017 “Microsoft Imagine Cup” Winner (Qatar round & MEA Round, Worldwide Finalist) 🔗

2017 “Best Bachelor Senior Project” Award, by Qatar National Research Fund (QNRF).

- 2017 Qatar University Awards for “Non-Academic Activities” and “Voluntary Activities”
2016 & 2017 First Place in The GCC Robotics Challenge - Qatar Round.

Talks

Tutorials

- 2024 “Optimistic Learning for Resource Management in Communication Systems” (Joint with G. Iosifidis) at ACM SIGMETRICS, Venice, Italy.

Seminar

- 2024 “Optimistic Learning”, ELLIS Network Seminars, TU Delft.
2023 “Adversarial Online Learning” at the Department of Computer Science & Engineering, Qatar University.

Conferences

- 2026 Poster presentation at *AISTATS*, Tangier, Morocco.
2025 Poster presentation at *ICML*, Vancouver, Canada.
2024 Presentation at *IEEE CDC*, Milan, Italy.
2024 Poster presentation at *L4DC*, University of Oxford, UK.
2022 Presentation at *IFIP Networking*, University of Catania, Sicily, Italy.

Teaching

- 2024 **CSE 3000 Research Project, TU Delft.**
○ [Machine Learning Algorithms for Caching Systems](#) (Bachelor Thesis of R. Vadastreanu)
○ Daily supervision and participation in the grading.
- 2024 **CSE 3000 Research Project, TU Delft.**
○ [Meta-learning the Best Caching Expert](#) (Bachelor Thesis of M. de Vries)
○ Daily supervision and participation in the grading.
- 2024 **CSE 3000 Research Project, TU Delft.**
○ [Optimistic Discrete Caching with Switching Costs](#) (Bachelor Thesis of L. Tosa)
○ Daily supervision and participation in the grading.
- 2022 **CSE 3000 Research Project, TU Delft.**
○ [Integral Caching using Online Mirror Descent](#) (Bachelor Thesis of Q. Oschatz)
○ Daily supervision and participation in the grading.
- 2022 **IN 4390: Quantitative Evaluation of Embedded Systems, TU Delft.**
○ Conduct and run lab sessions for practical assignment and additional exercises.
○ Solve and Grade assignments.

Publication List

Peer-Reviewed Journal/Magazine Articles

- [J15] M. Helmy, A. A. Abdellatif, **N. Mhaisen**, A. Mohamed, A. Erbad. “Slicing for AI: An Online Learning Framework for Network Slicing Supporting AI Services”. In: *IEEE Transactions on Network and Service Management* (2025).
- [J14] N. Khial, M. S. Allahham, **N. Mhaisen**, L. Ismail, M. Mabrok, A. Mohamed. “DRONE-RL: Dynamic Reinforcement learning for Online Navigation of UAVs in Evolving environments”. In: *Knowledge-Based Systems* (2025).
- [J13] N. Khial, **N. Mhaisen**, M. Mabrok, A. Mohamed. “An online learning framework for UAV search mission in adversarial environments”. In: *Expert Systems with Applications* 267 (2025).
- [J12] A. A. Abdellatif, **N. Mhaisen**, A. Mohamed, A. Erbad, M. Guizani. “Reinforcement learning for intelligent healthcare systems: A review of challenges, applications, and open research issues”. In: *IEEE Internet of Things Journal* 10.24 (2023).

- [J11] **N. Mhaisen**, G. Iosifidis, D. Leith. "Online caching with no regret: Optimistic learning via recommendations". In: *IEEE Transactions on Mobile Computing* 23.5 (2023).
- [J10] **N. Mhaisen**, A. Sinha, G. Paschos, G. Iosifidis. "Optimistic no-regret algorithms for discrete caching". In: *Proceedings of the ACM on Measurement and Analysis of Computing Systems (SIGMETRICS)* 6.3 (2022).
- [J9] A. A. Abdellatif, **N. Mhaisen**, A. Mohamed, A. Erbad, M. Guizani, Z. Dawy, W. Nasreddine. "Communication-efficient hierarchical federated learning for IoT heterogeneous systems with imbalanced data". In: *Future Generation Computer Systems* 128 (2022).
- [J8] M. S. Allahham, A. A. Abdellatif, **N. Mhaisen**, A. Mohamed, A. Erbad, M. Guizani. "Multi-agent reinforcement learning for network selection and resource allocation in heterogeneous multi-RAT networks". In: *IEEE Transactions on Cognitive Communications and Networking* 8.2 (2022).
- [J7] E. Baccour, **N. Mhaisen**, A. A. Abdellatif, A. Erbad, A. Mohamed, M. Hamdi, M. Guizani. "Pervasive AI for IoT applications: A Survey on Resource-efficient Distributed Artificial Intelligence". In: *IEEE Communications Surveys & Tutorials* (2022).
- [J6] J. Zheng, K. Li, **N. Mhaisen**, W. Ni, E. Tovar, M. Guizani. "Exploring deep-reinforcement-learning-assisted federated learning for online resource allocation in privacy-preserving EdgeloT". In: *IEEE Internet of Things Journal* 9.21 (2022).
- [J5] **N. Mhaisen**, A. A. Abdellatif, A. Mohamed, A. Erbad, M. Guizani. "Optimal user-edge assignment in hierarchical federated learning based on statistical properties and network topology constraints". In: *IEEE Transactions on Network Science and Engineering* 9.1 (2021).
- [J4] **N. Mhaisen**, M. S. Allahham, A. Mohamed, A. Erbad, M. Guizani. "On designing smart agents for service provisioning in blockchain-powered systems". In: *IEEE Transactions on Network Science and Engineering* 9.2 (2021).
- [J3] **N. Mhaisen**, N. Fetais, A. Erbad, A. Mohamed, M. Guizani. "To chain or not to chain: A reinforcement learning approach for blockchain-enabled IoT monitoring applications". In: *Future Generation Computer Systems* 111 (2020).
- [J2] **N. Mhaisen**, Q. M. Malluhi. "Data consistency in multi-cloud storage systems with passive servers and non-communicating clients". In: *IEEE Access* 8 (2020).
- [J1] **N. Mhaisen**, N. Fetais, A. Massoud. "Secure smart contract-enabled control of battery energy storage systems against cyber-attacks". In: *Alexandria Engineering Journal* 58.4 (2019).

Peer-Reviewed Conference Papers

- [C14] **N. Mhaisen**, G. Iosifidis. "Partially Lazy Gradient Descent for Smoothed Online Learning". In: *Twenty-Ninth Annual Conference on Artificial Intelligence and Statistics (AISTATS)*. 2026.
- [C13] **N. Mhaisen**, G. Iosifidis. "On the Dynamic Regret of Following the Regularized Leader: Optimism with History Pruning". In: *Forty-second International Conference on Machine Learning (ICML)*. 2025.
- [C12] N. Khial, **N. Mhaisen**, M. Mabrok, A. Mohamed. "An Online Learning Framework for UAV Target Search Missions in Non-Stationary Environments". In: *2024 IEEE Canadian Conference on Electrical and Computer Engineering (CCECE)*. 2024.
- [C11] **N. Mhaisen**, G. Iosifidis. "Adaptive online non-stochastic control". In: *Annual Learning for Dynamics & Control Conference (L4DC)*. PMLR. 2024.
- [C10] **N. Mhaisen**, G. Iosifidis. "Optimistic Online Non-stochastic Control via FTRL". In: *Conference on Decision and Control (CDC)*. 2024.
- [C9] **N. Mhaisen**, A. Sinha, G. Paschos, G. Iosifidis. "Optimistic No-regret Algorithms for Discrete Caching". In: *Abstract Proceedings of the ACM SIGMETRICS Conference on Measurement and Modeling of Computer Systems*. 2023.
- [C8] J. Zheng, K. Li, **N. Mhaisen**, W. Ni, E. Tovar, M. Guizani. "Federated learning for online resource allocation in mobile edge computing: A deep reinforcement learning approach". In: *2023 IEEE Wireless Communications and Networking Conference (WCNC)*. 2023.
- [C7] **N. Mhaisen**, G. Iosifidis, D. Leith. "Online caching with optimistic learning". In: *IFIP Networking Conference (IFIP Networking)*. 2022.
- [C6] **N. Mhaisen**, A. Mohamed, A. Erbad, M. Guizani. "Rational Contracts: Data-driven Service Provisioning in Blockchain-powered Systems". In: *ICC-IEEE International Conference on Communications*. 2021.
- [C5] **N. Mhaisen**, N. Fetais, A. Massoud. "Real-time scheduling for electric vehicles charging/discharging using reinforcement learning". In: *2020 IEEE International Conference on Informatics, IoT, and Enabling Technologies (ICIoT)*. 2020.
- [C4] **N. Mhaisen**, O. Abazeed, Y. Al-Hariri, N. Nawaz, A. Amira. "A reconfigurable multipurpose system on chip platform for metal detection". In: *IEEE 10th GCC Conference & Exhibition (GCC)*. 2019.
- [C3] O. Al Rshid Abazeed, **N. Mhaisen**, Y. Al-Hariri, N. Nawaz, A. Amira. "A Reconfigurable Multipurpose SoC Mobile Platform for metal detection". In: *Qatar Foundation Annual Research Conference Proceedings*. 2018.

- [C2] **N. Mhaisen**, O. Abazeed, Y. Al Hariri, A. Alsalemi, O. Halabi. "Self-powered IoT-enabled water monitoring system". In: *International Conference on Computer and Applications (ICCA)*. 2018.
- [C1] **N. Mhaisen**, M. Punkar, Y. Wang, Y. Desmedt, Q. Malluhi. "Almost BPXOR Coding Technique for Tolerating Three Disk Failures in RAID7 Architectures". In: *Qatar Foundation Annual Research Conference Proceedings*. 2018.

Patents

- [P2] Q. Malluhi, **N. Mhaisen**. "Data storage methods and systems". US Patent 11,748,197. Sept. 2023.
- [P1] Q. Malluhi, **N. Mhaisen**. "Data storage methods and systems". US Patent 11,531,647. Dec. 2022.

Preprints

- [A2] G. Iosifidis, **N. Mhaisen**, D. Leith "Optimistic Learning for Communication Networks" arXiv 2504.03499. *Monograph accepted in Foundation and Trends (FnT) in Networking, expected publication Q3 2026.*
- [A1] **N. Mhaisen**, G. Iosifidis "Dynamic Regret for Online Convex Optimization with Indicator Switching Costs". *Submitted to NeurIPS 2026.*